

Question:

Assume that B Corporation has the following total profit function equation: Total Profit =  $\pi = - \$3,000 - \$2,400 Q + \$350Q^2 - \$8.333Q^3$  a) Compute the marginal profit b) Determine its output quantities

Answer Choices:

- A. 22 and 7 units
- B. 21 and 8 units
- C. 23 and 6 units
- D. 26 and 2 units
- E. 24 and 4 units
- F. 25 and 3 units
- G. 20 and 5 units
- H. 27 and 1 unit
- I. 18 and 10 units
- J. 19 and 9 units

Reasoning + Answer:

<reasoning process here>

We are given the total profit function :

$$\pi = -3000 - 2400Q + 350Q^2 - 8.333Q^3$$

a) Compute the marginal profit .

Marginal profit ( MP ) is the derivative of total profit with respect to Q :

$$MP = d\pi/dQ$$

Differentiate term by term :

$$\begin{aligned} -d/dQ(-3000) &= 0 \\ -d/dQ(-2400Q) &= -2400 \\ -d/dQ(350Q^2) &= 700Q \\ -d/dQ(-8.333Q^3) &= -25Q^2 \end{aligned}$$
Wait â€“ actually :

$$d/dQ(-8.333Q^3) = -8.333 \times 3Q^2 = -25Q^2 \text{ (since } 8.333 \times 3 = 25 \text{)}$$

So :

$$MP = -2400 + 700Q - 25Q^2$$

Therefore , marginal profit function is :

$$MP = -2400 + 700Q - 25Q^2$$

b) Determine its output quantities .

The question says "determine its output quantities " â€“ likely meaning the quantity ( ies ) where marginal profit is zero ( since that 's typical for profit maximization ).

Set marginal profit equal to zero :

$$-2400 + 700Q - 25Q^2 = 0$$

Multiply both sides by -1 :

$$2400 - 700Q + 25Q^2 = 0$$

Rewriting :

$$25Q^2 - 700Q + 2400 = 0$$

Divide entire equation by 25 to simplify :

$$Q^2 - 28Q + 96 = 0$$

Now solve the quadratic equation :

$$\begin{aligned} Q &= [28 \pm \sqrt{(28^2 - 4 \times 1 \times 96)}] / 2 \\ &= [28 \pm \sqrt{(784 - 384)}] / 2 \\ &= [28 \pm \sqrt{400}] / 2 \\ &= [28 \pm 20] / 2 \end{aligned}$$

So :

$$\begin{aligned} Q &= (28 + 20) / 2 = 48 / 2 = 24 \\ Q &= (28 - 20) / 2 = 8 / 2 = 4 \end{aligned}$$

So the two output quantities are Q = 24 and Q = 4

Therefore , the pair is 24 and 4 units .

Looking at the answer choices , that matches :

E . 24 and 4 units

Note : The question says "output quantities " , plural â€“ and the two roots of the marginal profit function are at Q = 24 and Q = 4 â€“ likely meaning the profit-maximizing quantities .

Also , since marginal profit is a downward opening parabola , the maximum occurs at the vertex or where MP = 0 , and we have two roots â€“ so possibly two break-even or profit-extremum points â€“ but the typical interpretation for "determine its output quantities" in such problems is the values of Q where marginal profit is zero â€“ so we take both roots .

Thus , the correct answer is the pair ( 24 , 4 )

</reasoning process here>  
<answer>E</answer>